Advanced Public Economics — Applied Tutorial

Session III: Estimating Shape of Optimal Tax Rates

Daniel Weishaar

Roadmap for Session III

Estimate the **shape of optimal tax rates** over the income distribution.

1. Estimate income distributions in a non-parametric way (kernel density).
2. Add Pareto tail to the distribution.
3. Calibrate skill/ability from observed tax system.
4. Estimate shape of optimal tax rates.

Brief Recap of the Formula

Optimal Tax Rates — Based on Primitives

Following Diamond (1998), under the assumption of quasi-linear utility function with iso-elastic effort costs, i.e.,

$$u(c(\omega), y(\omega), \omega) = c(\omega) - k(y(\omega), \omega), \qquad k(y(\omega), \omega) = \left(1 + \frac{1}{\varepsilon}\right)^{-1} \left(\frac{y(\omega)}{\omega}\right)^{1+\frac{1}{\varepsilon}},$$

the **optimal non-linear tax rate** satisfies the condition

$$\frac{T'(y(\omega))}{1 - T'(y(\omega))} = (1 - G(\omega)) \frac{1 - F(\omega)}{f(\omega)\omega} \frac{1}{\varepsilon}$$

Types or productive abilities generally hard to observe, but we can rewrite the condition in terms of observable statistics, in particular the income distribution (Saez, 2001).

Optimal Tax Rates — Based on Observables

Based on monotonicity of $y(\omega)$, $F_y(y') = F(\omega_0(y'))$ and $f_y(y') = f(\omega_0(y'))\,\omega_0'(y')$.

Thus, the formula reads

$$\frac{T'(y)}{1 - T'(y)} = (1 - \mathcal{G}(y))\,\frac{1 - F_y(y)}{f_y(y)}\,\frac{\omega_0'(y)}{\omega_0(y)}\frac{1}{\varepsilon}$$

For now, further assume that exogenous welfare weights are Rawlsian, which yields the revenue-maximizing non-linear tax rate

$$\frac{T'(y)}{1 - T'(y)} = \frac{1 - F_y(y)}{f_y(y)}\,\frac{\omega_0'(y)}{\omega_0(y)}\frac{1}{\varepsilon}$$

Question: How do we bring this formula to the data?

- Requires income distribution, income-ability mapping, and behavioral responses.

Optimal Tax Rates — Inverse Hazard Rate

$$\frac{T'(y)}{1-T'(y)} = \frac{1-F_y(y)}{f_y(y)} \frac{\omega_0'(y)}{\omega_0(y)} \frac{1}{\varepsilon}$$

Inverse hazard rate requires estimation of cumulative distribution function (CDF) and probability density function (PDF). Different methods for estimation:

- **Parametric:** assume a functional form for the distribution (log-normal, Pareto,...) and select parameters to fit the data, e.g., through Maximum-Likelihood.
  - Example: We estimated the Pareto parameter for the top of the distribution.
  - But! Empirically observed distribution might feature irregular patterns that cannot be captured by one functional form.
- **Non-Parametric:** Start with the histogram of incomes and estimate the density using a **kernel density**. Intuitively, the kernel density estimator smooths the information from histogram of incomes.

Kernel Density Estimation

Optimal Tax Rates — Kernel Density Estimator

We follow the explanations provided in this data science blogpost, see also visualization.

Goal: Estimate an unknown PDF $f(x)$ without assuming a parametric form.

Key idea: Build the density from identical "building blocks" (kernels), one placed at each observation. Each block is a smooth PDF centered at a data point.

Optimal Tax Rates — Kernel Density Estimator, Toy Example

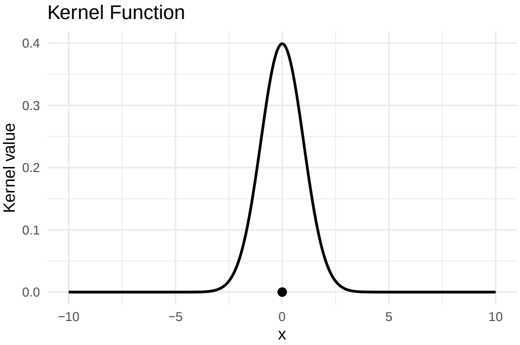
Start with a single kernel

Kernel is a valid probability density with unit area:

$$K(x) \geq 0, \qquad \int_{-\infty}^{\infty} K(x)\, dx = 1.$$

Example: Gaussian kernel

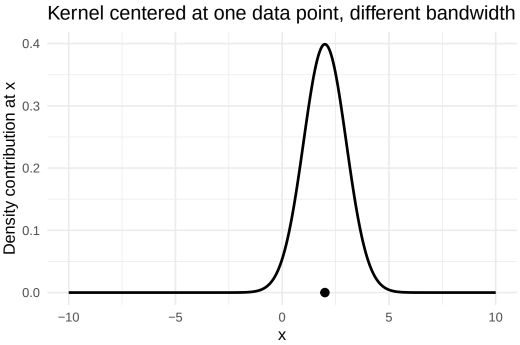
$$K(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$



Optimal Tax Rates — Kernel Density Estimator, Toy Example

Shift the kernel to a data point x_i:

$$K(x - x_i).$$



Optimal Tax Rates — Kernel Density Estimator, Toy Example

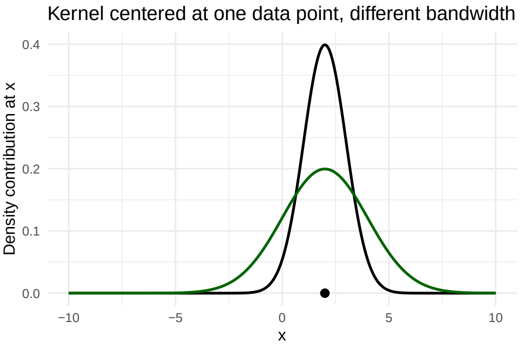
Control smoothness by scaling with the bandwidth *h*:

$$K\left(\frac{x - x_i}{h}\right).$$

Scaling changes the area, so we renormalize:

$$\frac{1}{h}K\left(\frac{x - x_i}{h}\right).$$

- large h: wide, smooth density;
- small h: narrow, spiky density.



Optimal Tax Rates — Kernel Density Estimator, Toy Example

Multiple data points For each data point x_i create a kernel contribution

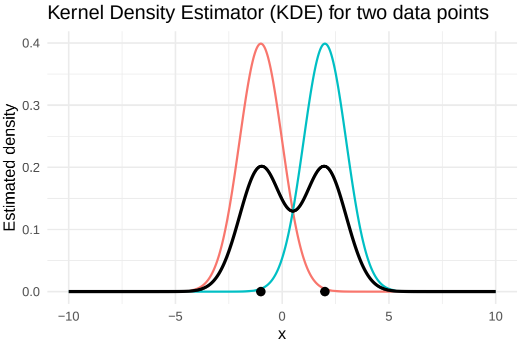
$$f_i(x) = \frac{1}{h} K\left(\frac{x - x_i}{h}\right).$$

Add all contributions:

$$\tilde{f}(x) = \sum_{i=1}^{n} f_i(x).$$

This has total area n. Normalize by dividing by n to obtain a proper PDF:

$$\widehat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right).$$



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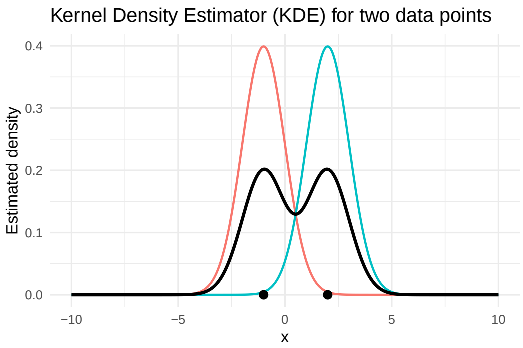

$\Rightarrow$ Hands-on exercise:

Optimal Tax Rates — Kernel Density Estimator — Specifications

The **Kernel Density Estimator (KDE)** is defined via

- **Kernel** — shape with which the pdf around each observation is approximated (e.g, Gaussian, uniform, triangular, biweight, triweight, Epanechnikov).

  $\Rightarrow$ Choice is of second-order importance.

- **Bandwidth** — degree of smoothing. Selecting bandwidth implies choosing how to resolve bias-variance trade-off. $\Rightarrow$ First-order importance.
  - **Small** $h \Rightarrow$ undersmoothing, high variance, many small peaks.
  - **Large** $h \Rightarrow$ oversmoothing, high bias, important features disappear.

Bandwidth Selection: Choose h to balance bias (too large h) and variance (too small h), i.e., we minimize mean integrated squared error (MISE) $\rightarrow$ optimal bandwidth h^*.

$$\text{MISE}(h) = \mathbb{E}\left[\int \left(\hat{f}_h(x) - f(x)\right)^2 dx\right].$$

Optimal Tax Rates — Kernel Density Estimator — Bandwidth Selection

1. **Rule-of-Thumb, see Silverman (1986)**
   - Assumes data come from a Gaussian distribution.
   - Optimal bandwidth is then function of observed characteristics of this distribution.
2. **Adaptive (Variable) Bandwidths**
   - Allow h_i to vary: small h_i in dense regions, large in sparse regions.
   - Useful for highly skewed or heavy-tailed data (e.g. income distributions).

(3. Cross-Validation Methods)

- Choose h by optimizing out-of-sample fit.

Implementation in R via `density` command. Focus on $1 + 2$ in hands-on exercise.

Optimal Tax Rates — KDE — Bandwidth — Silverman Rule

Assumption:

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Silverman (1986) shows that asymptotically **optimal bandwidth** (minimizing MISE) is

$$h_{\text{Silverman}} = 0.9 \min\left(\sigma, \frac{\text{IQR}}{1.34}\right) n^{-1/5},$$

where σ is the sample standard deviation and IQR is the inter-quartile range.

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Implementation: In R, use `density(x, bw = "nrd0")`

$\Rightarrow$ Good baseline, but often insufficient for income data which is skewed.

Optimal Tax Rates — KDE — Bandwidth — Variable Bandwidth (I)

Standard KDE uses a *global* bandwidth h, but income distributions are highly skewed. A single h cannot smooth the bottom and the top well at the same time.

Variable Bandwidth Selection based on Abramson (1982):

- Use a *pilot density* $\hat{f}_0(y)$ to detect sparse vs. dense regions.
- Assign a local bandwidth h_i to each observation.

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Implementation:

$$h_i = \lambda_i \, h_0, \qquad \lambda_i = \left(\frac{\bar{f}}{\hat{f}_0(y_i)} \right)^{1/2},$$

where $\bar{f}$ is geometric mean of $\hat{f}_0(y_i)$. Exponent gives asymptotic variance stabilization.

Optimal Tax Rates — KDE — Bandwidth — Variable Bandwidth (II)

Adaptive KDE:

$$\hat{f}(y) = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{h_i}K\left(\frac{y-y_i}{h_i}\right).$$

Pilot density $\hat{f}_0$ selected via standard KDE, e.g. Silverman's rule of thumb.

Key point: More smoothing in sparse (top-income) regions, less smoothing in dense areas where we have more reliable information.

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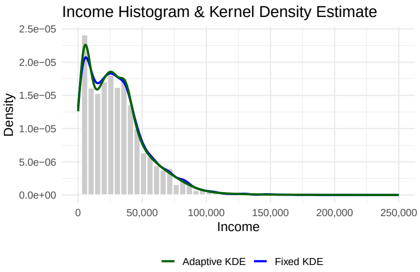
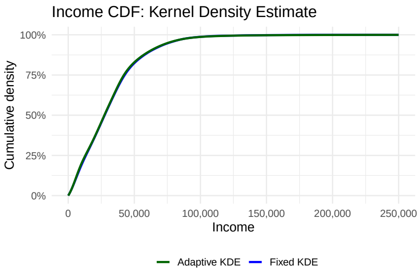
$\Rightarrow$ Hands-on exercise

Optimal Tax Rates — Inverse Hazard Rate

- We estimated pdf and cdf based on the SOEP training data.

  $\Rightarrow$ Difference between fixed and variable bandwidth seem small.





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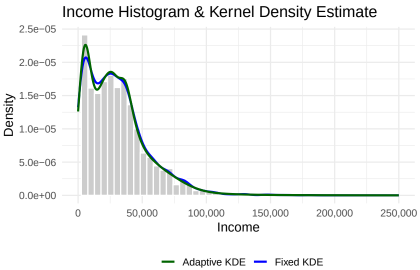

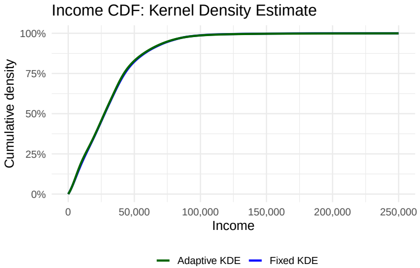

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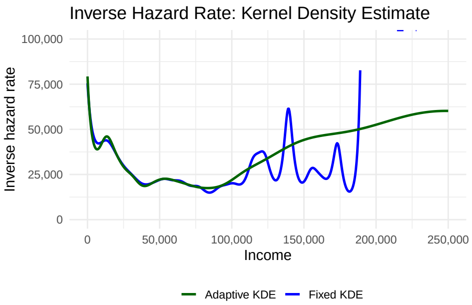
  $\Rightarrow$ Difference between fixed and variable bandwidth is substantial, because small local noise is magnified.



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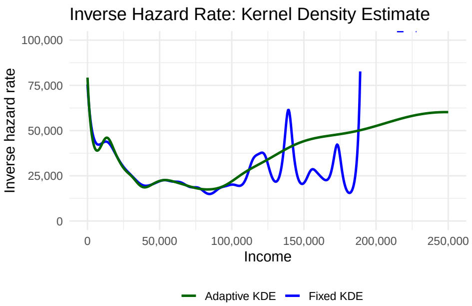

- Variable bandwidth KDE smoother at the top, but still not reliable.

$\Rightarrow$ Need to **append Pareto tail** to top!

Adding Pareto Tail

Optimal Tax Rates — Inverse Hazard Rate — Appending Pareto Tail (I)

Idea:

- Estimate the distribution with KDE up to y_{start}, e.g., top 3%.
- Transition from KDE to Pareto between y_{start} and y_{const}, e.g., top 1 %.
- Above y_{const}, distribution is Pareto with a_{const}.

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Procedure:

① Start transition at y_{start} and obtain a_{start} from $a_{\text{start}} = \frac{f^{KDE}(y_{\text{start}})}{1-F^{KDE}(y_{\text{start}})} \cdot y_{\text{start}}$.

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   3.1 Obtain y_{min} from y and a using $y_{\text{min}} = \left(\frac{f(y_i)\, y_i}{a_i}\right)^{\frac{1}{a_i}} y_i$.

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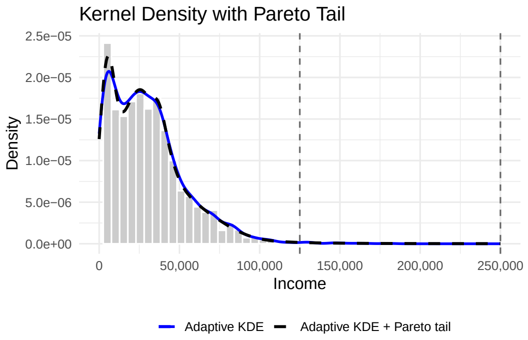
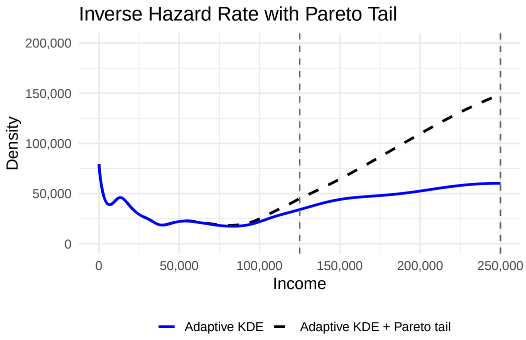
④ Finally, re-normalize so that $\int_0^\infty f(y)\, dy = 1$.

Optimal Tax Rates — Inverse Hazard Rate — Appending Pareto Tail (II)

Results:





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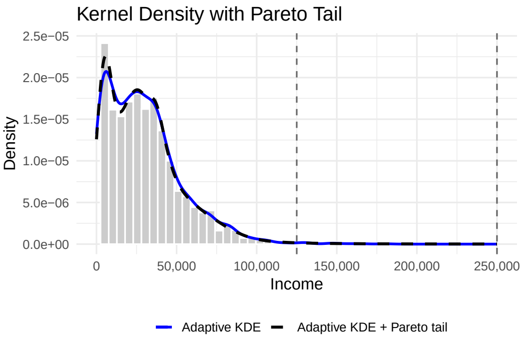

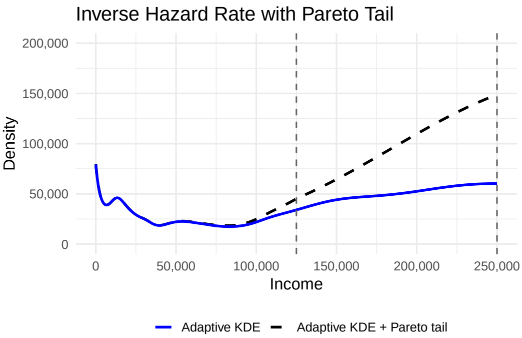

$\Rightarrow$ We have a reasonable estimate for the inverse hazard rate that combines non-parametric KDE with Pareto tail estimation at the top.

Skill/Ability Calibration

Optimal Tax Rates — Skill/Ability

Going back to the formula for optimal tax rates

$$\frac{T'(y)}{1 - T'(y)} = \frac{1 - F_y(y)}{f_y(y)} \frac{\omega_0'(y)}{\omega_0(y)} \frac{1}{\varepsilon}$$

we also need information on the ability/skill level of individuals.

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- Income per unit of effort $\omega = \frac{y}{l}$ would require measuring effort l. Could use working hours, but many unobservable effort components not captured in hours.

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- Income per unit of effort $\omega = \frac{y}{l}$ would require measuring effort l. Could use working hours, but many unobservable effort components not captured in hours.
- Thus, we usually consider ω to be unobserved.
- However, we can make use of the fact that individual utility maximization gives relation between ω and y, see Saez (2001).

Optimal Tax Rates — Skill/Ability — Inverting the FOC

Individuals choose income y given ability ω and marginal tax rate $T'(y)$. With iso-elastic effort costs, the first-order condition is

$$(1 - T'(y)) = \left(\frac{y}{\omega}\right)^{1/\varepsilon}.$$

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Inverting the FOC, we obtain

$$\omega(y) = \left(\frac{y^{1/\varepsilon}}{1 - T'(y)}\right)^{\frac{1}{1+1/\varepsilon}}.$$

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This requires

- information on y (observed),
- assumptions about the elasticity value ϵ,
- information on the status quo marginal tax rates $T'(y)$.

Optimal Tax Rates — Skill/Ability — Status-Quo Tax System (I)

To back out abilities, we need the *status-quo marginal tax rate* $T'(y)$. Statutory MTR obtained from tax code available here for 2019. $\Rightarrow$ Hands-on exercise.

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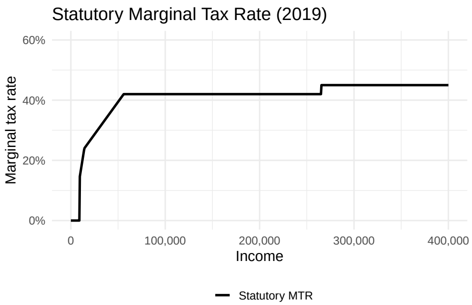
German tax schedule (2019)

- Tax-free allowance of 9,169 EUR.
- Two income ranges where MTR is linearly increasing (*Progressionszone*)
- *Spitzensteuersatz* of 42%.
- *Reichensteuersatz* of 45%.



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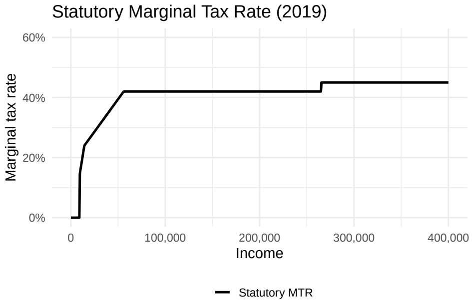

But careful! Kinks imply that the FOC does not identify the global maximum.

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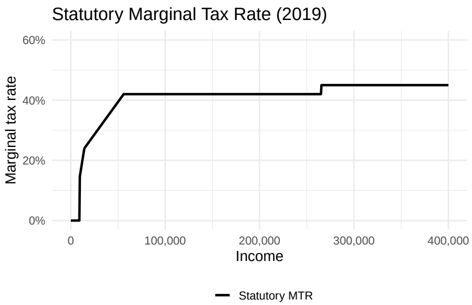
- *Reichensteuersatz* of 45%.



But careful! Kinks imply that the FOC does not identify the global maximum.

$\Rightarrow$ To make our lives simpler, we approximate tax function.

Optimal Tax Rates — Skill/Ability — Status-Quo Tax System (II)

We use a progressive function with constant rate of progressivity following Heathcote et al. (2017) (HSV):

$$c(y) = \lambda y^{1-\tau} \quad \Rightarrow \quad T(y) = y - \lambda y^{1-\tau}.$$

Estimate (λ, τ) from

$$\log c = \underbrace{\log \lambda}_{\alpha} + \underbrace{(1-\tau)}_{\beta} \log y.$$

This yields a smooth marginal tax rate:

$$T'(y) = 1 - \lambda(1-\tau)y^{-\tau}.$$

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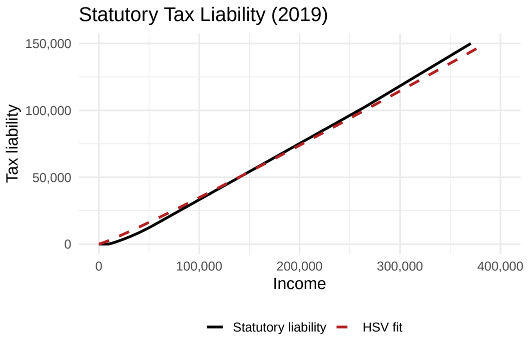
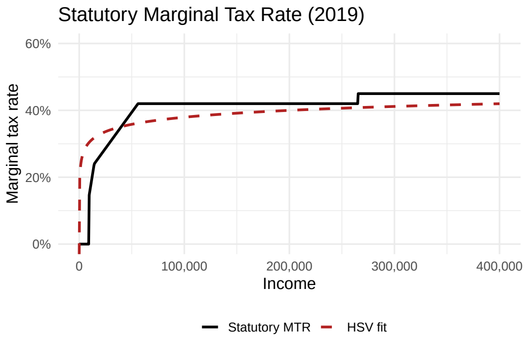
$\Rightarrow$ Hands-on exercise

Optimal Tax Rates — Skill/Ability — Status-Quo Tax System (III)

Result: Approximation preserves average progressivity, not exact location of kinks.





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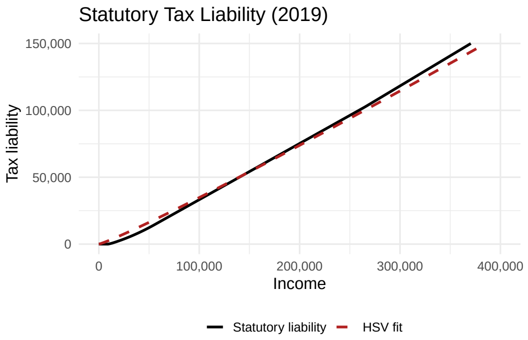
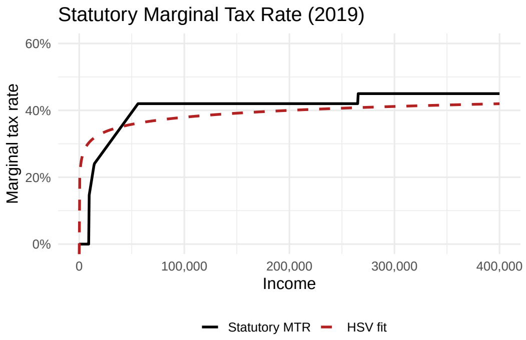




⇒ We now have all ingredients to estimate the shape of optimal tax rates!

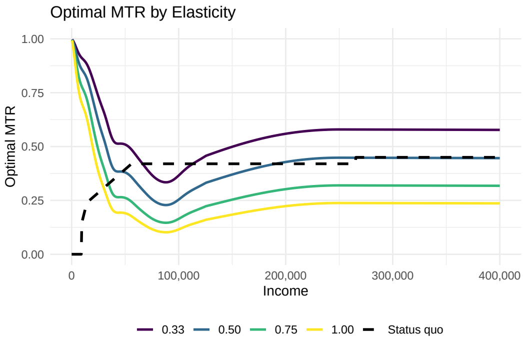
Shape of Optimal Tax Rates

Optimal Tax Rates — Resulting Optimal MTR

Combining all components, the optimal MTR can be estimated

$$\frac{T'(y)}{1-T'(y)} = \frac{1-F_y(y)}{f_y(y)} \cdot \frac{\omega'(y)}{\omega(y)} \cdot \frac{1}{\varepsilon}.$$

Note: $\omega'(y)$ can be estimated by numerical differentiation, i.e., $\frac{\Delta\omega}{\Delta y}$.



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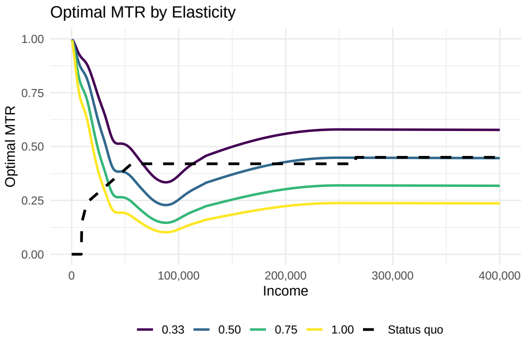
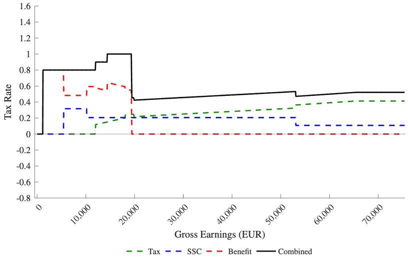


Observations:

- Well-known u-shape of optimal marginal tax rates.
- Under plausible elasticity of 0.5, optimal top tax rate fine, but starts too early.
- What about bottom tax rates?

Optimal Tax Rates — Resulting Optimal MTR — Remark

- For simplicity, we used **statutory MTR** of the income tax code.
- This does not account for the full tax system, in particular it misses **transfer withdrawal rate**, i.e., reduction in transfer payments once income goes up.
- The **effective MTR** has much higher values at the bottom.



Source: Hansen and Weishaar (2025)

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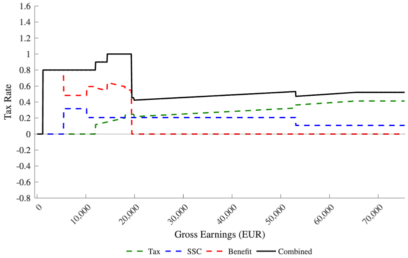

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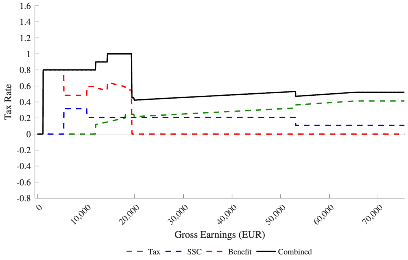

Source: Hansen and Weishaar (2025)

$\Rightarrow$ At first glance, effective MTR seems more in line with optimal shape.
$\Rightarrow$ However, as we will see in the next sessions, this is not the case.

- Preview: the optimal shape of marginal tax rates is smoothly decreasing while the effective MTR features discrete drops.

That's it! See you next time!

References

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